

Assignment No 22

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Domain : AI & ML

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Q1.

```
print(shape of y, y.shape)
```

... First 5 Rows:

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	\
0	40	M	ATA	140	289	0	Normal	172	
1	49	F	NAP	160	180	0	Normal	156	
2	37	M	ATA	130	283	0	ST	98	
3	48	F	ASY	138	214	0	Normal	108	
4	54	M	NAP	150	195	0	Normal	122	

	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	N	0.0	Up	0
1	N	1.0	Flat	1
2	N	0.0	Up	0
3	Y	1.5	Flat	1
4	N	0.0	Up	0

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 918 entries, 0 to 917

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	Age	918 non-null	int64
1	Sex	918 non-null	object
2	ChestPainType	918 non-null	object
3	RestingBP	918 non-null	int64
4	Cholesterol	918 non-null	int64
5	FastingBS	918 non-null	int64
6	RestingECG	918 non-null	object
7	MaxHR	918 non-null	int64
8	ExerciseAngina	918 non-null	object

```

6  RestingECG      918 non-null  object
7  MaxHR           918 non-null  int64
8  ExerciseAngina  918 non-null  object
...
9  Oldpeak         918 non-null  float64
10 ST_Slope        918 non-null  object
11 HeartDisease    918 non-null  int64
dtypes: float64(1), int64(6), object(5)
memory usage: 86.2+ KB
None

```

Encoded Feature Columns:

```

Index(['Age', 'RestingBP', 'Cholesterol', 'FastingBS', 'MaxHR', 'Oldpeak',
       'Sex_M', 'ChestPainType_ATA', 'ChestPainType_NAP', 'ChestPainType_TA',
       'RestingECG_Normal', 'RestingECG_ST', 'ExerciseAngina_Y',
       'ST_Slope_Flat', 'ST_Slope_Up'],
      dtype='object')

```

First 5 Rows of Encoded Features:

	Age	RestingBP	Cholesterol	FastingBS	MaxHR	Oldpeak	Sex_M \
0	40	140	289	0	172	0.0	True
1	49	160	180	0	156	1.0	False
2	37	130	283	0	98	0.0	True
3	48	138	214	0	108	1.5	False
4	54	150	195	0	122	0.0	True

	ChestPainType_ATA	ChestPainType_NAP	ChestPainType_TA	RestingECG_Normal \
0	True	False	False	True
1	False	True	False	True
2	True	False	False	False
3	False	False	False	True
4	False	True	False	True

RestingECG_ST ExerciseAngina_Y ST_Slope_Flat ST_Slope_Up

```

4         false         true         false         true
      RestingECG_ST  ExerciseAngina_Y  ST_Slope_Flat  ST_Slope_Up
0         False         False         False         True
1         False         False         True         False
2          True         False         False         True
3         False         True         True         False
4         False         False         False         True

```

Target Variable (y):

```

0    0
1    1
2    0
3    1
4    0

```

Name: HeartDisease, dtype: int64

Shape of X: (918, 15)

Shape of y: (918,)

Q2.

```
*** X_train shape: (734, 15)
    X_test shape : (184, 15)
    y_train shape: (734,)
    y_test shape : (184,)
```

Q3.

▼ ... Logistic Regression model trained successfully!
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (50 iterations total). Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(

Q4.

```
*** First 10 Actual Values:
668    0
30     1
377    1
535    1
807    0
793    1
363    1
583    0
165    1
483    1
Name: HeartDisease, dtype: int64

First 10 Predicted Values:
[0 0 1 1 0 1 1 0 1 1]
```

Q5.

```
*** Confusion Matrix:
[[66 11]
 [17 90]]

Confusion Matrix Labels:
True Negative (TN): 66
False Positive (FP): 11
False Negative (FN): 17
True Positive (TP): 90
```


Q6.

```
*** Accuracy : 0.8478260869565217
Precision: 0.8910891089108911
Recall   : 0.8411214953271028
F1 Score : 0.8653846153846154
```

Classification Report:

	precision	recall	f1-score	support
0	0.80	0.86	0.82	77
1	0.89	0.84	0.87	107
accuracy			0.85	184
macro avg	0.84	0.85	0.85	184
weighted avg	0.85	0.85	0.85	184

Q7.

```
[25]
✓ Os 
# Import joblib
import joblib

joblib.dump(model, "heart_model.pkl")
joblib.dump(scaler, "scaler.pkl")
joblib.dump(X.columns.tolist(), "columns.pkl")

# Display success message
print("Model and preprocessing objects saved successfully!")

▼ ... Model and preprocessing objects saved successfully!
```

Q8.

```
print("Result: No Heart Disease")
```

```
*** Prediction: 0  
Result: No Heart Disease
```

Q9.

```
2026-07-21 14:46:32.788 WARNING streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread':
2026-07-21 14:46:32.797 WARNING streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread':
2026-07-21 14:46:33.303
```

Warning: to view this Streamlit app on a browser, run it with the following command:

```
streamlit run /usr/local/lib/python3.12/dist-packages/colab_kernel_launcher.py [ARGUMENTS]
2026-07-21 14:46:33.305 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.307 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.310 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.313 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.322 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.337 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.340 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.346 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.352 Session state does not function when running a script without `streamlit run`
2026-07-21 14:46:33.358 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.365 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.368 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.375 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.385 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.390 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.394 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.401 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.406 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.410 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.417 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
2026-07-21 14:46:33.418 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when runni
```


[illegible]